

Supplementary Materials

BCHAIN-ACCESS: A Standards-Informed, Evidence-Aware Framework for Accessibility-Oriented Blockchain Healthcare Consent Interfaces

This supplement provides implementation-depth and audit material referenced from the main paper. Following the main paper, **Cycle 1** denotes the initial prototype evaluated by the ten experts and **Cycle 2** the refined artifact verified on 15 July 2026 at commit `360eaa2a26b2f2b4fef98c0667f87db09e5e5120`. A 19 July text-only working-tree correction aligns the review-screen duration with the gateway’s no-expiry default and is not included in the 15 July accessibility evidence. Evidence is reported against the cycle that produced it. S1 reports development-network measurements from a local Hardhat EVM node. S2–S4 document implementation assumptions, conceptual scope distinctions, security controls, and residual risks. These materials are not representative of production-network behavior; they record tested evidence, scope, and residual risks.

S1. Final-Contract Development-Network Gas Results

Regenerated on 15 July 2026 from the final Cycle 2 contract at code/evidence commit `360eaa2a26b2f2b4fef98c0667f87db09e5e5120`, using the complete 28-test workload, solc 0.8.26, optimizer enabled (200 runs), Paris EVM target, Hardhat chain ID 1337, and a block limit of 30,000,000 gas. Values are development-node `gasUsed`; fee conversions are omitted because the intended setting is a permissioned consortium network.

Contract Function	EVM Gas	Interpretation
<code>registerRecord</code>	341,833 ($n = 28$)	Write (storage initialization plus audit entry)
<code>initiateGrant</code>	335,903	Write (229,466–373,149; $n = 18$)
<code>confirmGrant</code>	200,852 ($n = 10$)	Modify (storage update plus audit entry)
<code>cancelPendingGrant</code>	213,322 ($n = 6$)	Modify (pending grant cleared plus audit entry)
<code>revokeAccess</code>	173,117	Modify (171,917–173,917; $n = 5$)
<code>logAccessAttempt</code>	176,054	Write (175,032–177,076; $n = 4$)
<code>setAuthorizedGateway</code>	47,823 ($n = 3$)	Administrative gateway update
<code>pause</code>	45,540 ($n = 5$)	Administrative emergency-state update
<code>unpause</code>	23,634 ($n = 4$)	Administrative emergency-state update
<code>transferOwnership</code>	28,600 ($n = 2$)	Modify (owner update plus event)
<code>checkAccess</code>	Not measured by transaction	No user-paid transaction gas when invoked off-chain through <code>eth_call</code> ; an on-chain invocation would consume gas.
<code>getAuditLog</code>	Not measured by transaction	No user-paid transaction gas when invoked off-chain through <code>eth_call</code> ; an on-chain invocation would consume gas.
<i>(deployment)</i>	2,934,432	Deployment (9.8% of the development block limit)

All 28 tests passed during this gas-reporting run. The retained report is `results/gas-report.txt`. For an identical EVM revision, contract bytecode, input, and storage state, `gasUsed` is deterministic. Effective fees, congestion, RPC behavior, confirmation time, and network-specific execution conditions were not evaluated. The demonstration registered seeded 32-byte identifiers; it did not include clinical record files or a production off-chain store.

S2. Assumption–Threat–Control–Residual-Risk Matrix

Assumption	Threat	Implemented control	Residual risk	Not evaluated
Patient key is controlled by the patient	Unauthorized grant	onlyPatient and record-owner mapping	Key theft, malicious wallet, and identity mismatch	HSM, recovery, identity proofing
EVM transaction ordering is valid	Replay	Account nonces	Application-layer replay and cross-domain signing	Adversarial replay tests
Patient can act without coercion	Accidental or coerced grant	Minimum wait and cancellation until confirmation	Coercer can wait; control mitigates accidents only	Coercion-resistance study
Chain history remains available	Log alteration	Events and append-only contract array	Gateway logs, reorgs, governance, and display integrity	End-to-end audit assurance
Addresses correspond to approved actors	Sybil identities	Application provider registry	Registry is not proof of personhood or organizational identity	Identity governance
Only the patient confirms	Ordering/front-running	Caller restriction and explicit provider/record tuple	MEV and transaction-ordering risks remain	Public-testnet/MEV analysis
REST request contains multiple records	Partial multi-record commit	The final gateway rejects any state-changing consent request that does not contain exactly one record identifier	No atomic multi-record API is provided	Batch-contract design and gateway failure-injection tests if batch support is introduced
Access-attempt logger is called	Misleading or spam audit entries	Contract restricts logging to the owner or configured authorized gateway and retains the caller address	Gateway key compromise, role governance, and rate limiting remain deployment risks	Adversarial authorization and rate-limit testing
Emergency pause is invoked	Continued grant creation or confirmation during an incident	whenNotPaused applies to grant initiation and confirmation; cancellation and revocation remain available	Operational pause governance and key compromise remain unassessed	Incident-response and adversarial pause testing
Known Hardhat accounts are used	Development keys reused outside the isolated node	Configuration labels the fallback key development-only and production mode requires an environment value	Provider/demo keys remain embedded for local demonstration	HSM/KMS-backed production key custody and rotation

S3. Conceptual Scope Distinctions Across Architecture Patterns

The centralized-EHR and pure-Ethereum columns are generic architecture categories, not evaluated comparator systems. The table presents conceptual distinctions only and does not provide empirical performance, privacy, usability, or accessibility comparisons.

Property	Centralized EHR	Pure-Ethereum	BCHAIN-ACCESS
Patient control over grants	Deployment-dependent	Deployment-dependent	Implemented for a reference patient address in the local authorization model; no user-controlled wallet integration
Personal-data exposure	Database-dependent	Design-dependent	Seeded 32-byte identifiers on-chain in the reference artifact; production hashes or metadata would require separate privacy engineering and off-chain storage
Delayed confirmation workflow	Not assessed	Not assessed	Implemented in reference contract
Audit evidence	Database logs	Ledger events	Contract events plus residual end-to-end risks
WCAG 2.2 inspection	Not assessed	Not assessed	Selected Cycle 1 issues addressed and technically verified across 12 Cycle 2 interface states; residual criterion-level and target-user questions remain; no conformance claim

S4. Security Threat Model (STRIDE)

STRIDE decomposition of `HealthAccessControl.sol`. “Residual/next step” items are scoped to the deployment, adversarial-testing, and independent-assurance work identified in Limitation L9.

STRIDE category	Contract-level exposure	Mitigation / residual
Spoofing (identity)	Impersonating a patient or provider address	EVM account keys + <code>onlyPatient(msg.sender)</code> binding; residual: key-management/HSM out of scope
Tampering (data)	Altering records or grants	On-chain seeded-identifier registration and append-only contract structures; clinical-record hash anchoring and production off-chain encryption were not demonstrated
Repudiation	Denying a grant/revoke occurred	Contract events and <code>AuditEntry[]</code> provide chain-scoped evidence; end-to-end attribution and display integrity remain unassessed
Information disclosure	Leaking PII via chain data	Demonstration registers seeded 32-byte identifiers only; real metadata may remain identifying, and no production encryption or salt store is implemented
Denial of service	Blocking access or exhausting gas	Off-chain <code>eth_call</code> reads incur no user-paid transaction gas; owner pause blocks new initiation and confirmation while cancellation/revocation remain available. Public-network fees, RPC availability, and denial-of-service behavior are unmeasured (L9)
Elevation of privilege	Gaining access or writing misleading audit entries	<code>onlyPatient</code> restrictions, delayed confirmation, and owner/authorized-gateway logger restriction; Cycle 2 bytecode received fresh Slither analysis but still requires adversarial testing, fuzzing, and independent audit

S5. WCAG 2.2 A/AA Criterion-Level Inspection Matrix

The matrix contains separate rows for every WCAG 2.2 Level A and AA success criterion. SC 4.1.1 Parsing is not included because it is obsolete and removed in WCAG 2.2. Each Cycle 1 result uses one of four classifications: *No issue observed*, *Issue observed*, *Not fully evaluated*, or *Not applicable*. “No issue observed” is limited to the retained method and states and is not a conformance finding. The **Cycle 1 tested state** column describes the build evaluated by the ten experts. State labels A1–A12 denote the Cycle 2 states enumerated in S5.2 and exist only in Cycle 2; they therefore appear in the method/provenance and Cycle 2 columns and never as the basis of a Cycle 1 result.

Evidence layer	Scope	Interpretation
Cycle 1 build	Enumerated workflow states used by the ten-expert assessment; commit and environment not archived (L4)	Cycle 1 manual results remain the evaluation evidence and are not replaced by Cycle 2 changes.
Cycle 2 refinement	Repository source after the Cycle 1 sessions	The final column records targeted source changes, subsequent technical-verification status, and remaining criterion-specific gaps; it is not a second conformance evaluation.
Cycle 2 scan, 15 July 2026	Initial rendered page only	Lighthouse 13.4.0: 100/100, 23 passed, 0 failed, 43 not applicable, 10 manual; axe-core 4.12.1: zero violations and zero incomplete. These automated results do not supersede the Cycle 1 manual findings.
Intermediate workflow inspection, 15 July 2026	Selected workflow and open audit-dialog state before the final dialog fix	Live axe-core 4.8.2 reported one moderate region finding. This result is retained as intermediate Cycle 2 evidence.
Final state-by-state verification, 15 July 2026	Twelve enumerated Cycle 2 states plus separate keyboard, focus, dialog, target-size, error-presentation, and reflow checks	Standalone axe-core 4.10.0 recorded zero violations in all 12 retained reports; five reports contain incomplete items retained for manual review. No conformance determination or live screen-reader test was performed.

S5.1 Deposited Automated Report Summary

Artifact	Date/build status	Tool-native result	Interpretation
Lighthouse, Cycle 1	12 July 2026	100/100; 23 passed, 0 failed, 43 not applicable, 10 manual	Not linked to a specific session build
Lighthouse, Cycle 2	15 July 2026	100/100; 23 passed, 0 failed, 43 not applicable, 10 manual	Initial-state Cycle 2 scan
WAVE	29 June 2026; retained URL	0 errors, 0 contrast errors, 0 alerts	Tool-limited retained report
Undated axe-core summary	Unknown	Three passed rule summaries	Incomplete evidence
axe-core, Cycle 2	15 July 2026	0 violations, 0 incomplete	Initial-state Cycle 2 scan
Live axe-core workflow/dialog inspection	15 July 2026; Cycle 2	One moderate region finding involving audit-dialog content	Targeted live-state inspection; not a complete conformance determination
Final standalone axe-core state reports	15 July 2026; final Cycle 2 build	0 violations across all 12 retained reports; 5 reports contain incomplete items	Final Cycle 2 engineering verification; manual checks recorded separately; no conformance determination

These artifacts differ in provenance, date, tool version, and retained state coverage and are not aggregated. Lighthouse 13.4.0 reports its own 100/100 composite and embedded axe-core 4.12.1 environment. The final state files are standalone axe-core 4.10.0 reports and are reported independently.

S5.2 Final State-by-State Evidence Inventory

State	Interface state	Retained files	Standalone axe result	Recorded manual/browser method and result	Evidence boundary
A1	Landing	A1_landing_axe.json; A1_landing.png	0 violations; 0 incomplete	Keyboard entry and visible focus inspected	No live screen-reader confirmation
A2	Dashboard	A2_dashboard_axe.json; A2_dashboard.png	0 violations; 0 incomplete	Programmatic dashboard-heading focus inspected	Cycle 2 build only
A3	Record selection	A3_record-selection_axe.json; A3_record-selection.png	0 violations; 0 incomplete	Provider selection, single-record choice, names, and target dimensions inspected	No alternative-input test
A4	Review	A4_review_axe.json; A4_review.png	0 violations; 0 incomplete	Wizard-transition focus and keyboard controls inspected	Comprehension effectiveness not evaluated
A5	Countdown	A5_countdown_axe.json; A5_countdown.png	0 violations; 1 incomplete	Focus order and status messaging inspected under SC 2.4.3 and SC 4.1.3; the minimum wait is not classified under SC 2.2.1	Acceptability of delay unresolved
A6	Confirmation success	A6_confirmation-success_axe.json; A6_confirmation-success.png	0 violations; 0 incomplete	Success focus and visible/programmatic status inspected	No live AT announcement confirmation
A7	Audit dialog open	A7_audit-dialog-open_axe.json; A7_audit-dialog-open.png	0 violations; 1 incomplete	Dialog name, focus containment, Escape handling, and restoration inspected	Manual browser result; no live screen reader
A8	Revocation dialog open	A8_revocation-dialog-open_axe.json; A8_revocation-dialog-open.png	0 violations; 1 incomplete	Dialog focus containment, Escape handling, cancellation, and restoration inspected	No live screen reader
A9	Expanded FAQ	A9_expanded-faq_axe.json; A9_expanded-faq.png	0 violations; 1 incomplete	Accordion keyboard state and affected target dimensions inspected	Content comprehension not evaluated
A10	Record-loading failure	A10_record-loading-failure_axe.json; A10_record-loading-failure.png	0 violations; 0 incomplete	Verification harness invoked the implemented visible alert and programmatic announcement presentation	Presentation check, not RPC failure injection
A11	Audit-loading failure	A11_audit-loading-failure_axe.json; A11_audit-loading-failure.png	0 violations; 1 incomplete	Verification harness invoked the implemented audit-dialog error presentation and focus movement	Presentation check, not RPC failure injection
A12	Transaction failure	A12_transaction-failure_axe.json; A12_transaction-failure.png	0 violations; 0 incomplete	Verification harness invoked the implemented visible alert and programmatic announcement presentation	Presentation check, not adversarial transaction testing

SC	Criterion	Applicability	Cycle 1 tested state	Method and state provenance	Cycle 1 result	Cycle 2 status
1.1.1	Non-text Content (A)	Applicable	Images, icons, and controls in enumerated states	Code and rendered-state inspection	No issue observed: retained inspection recorded text alternatives or decorative treatment.	No targeted source change; complete-process re-evaluation not performed.
1.2.1	Audio-only and Video-only (Prerecorded) (A)	Not applicable	All enumerated states	Content inventory	Not applicable: no prerecorded audio-only or video-only content.	No media added; applicability must be reconfirmed for future builds.
1.2.2	Captions (Prerecorded) (A)	Not applicable	All enumerated states	Content inventory	Not applicable: no prerecorded synchronized media.	No media added; applicability must be reconfirmed for future builds.
1.2.3	Audio Description or Media Alternative (Prerecorded) (A)	Not applicable	All enumerated states	Content inventory	Not applicable: no prerecorded video content.	No media added; applicability must be reconfirmed for future builds.
1.2.4	Captions (Live) (AA)	Not applicable	All enumerated states	Content inventory	Not applicable: no live synchronized media.	No live media added; applicability must be reconfirmed for future builds.
1.2.5	Audio Description (Prerecorded) (AA)	Not applicable	All enumerated states	Content inventory	Not applicable: no prerecorded video content.	No media added; applicability must be reconfirmed for future builds.
1.3.1	Info and Relationships (A)	Applicable	Page regions, forms, records, and grant wizard in the evaluated build	Code inspection (Cycle 1); standalone axe-core and manual dialog inspection in Cycle 2 state A7	No issue observed: retained review recorded semantic HTML and ARIA landmarks.	Remediated and technically verified in the final Cycle 2 build; <code>A7_audit-dialog-open_axe.json</code> records zero violations and the corresponding screenshot and manual dialog check are retained.
1.3.2	Meaningful Sequence (A)	Applicable	Reading order across enumerated states	DOM and manual reading-order inspection	No issue observed: retained review recorded logical DOM order.	No targeted change; complete-process re-evaluation not performed.
1.3.3	Sensory Characteristics (A)	Applicable	Instructions and state cues	Content inspection	No issue observed: instructions were not recorded as color-, shape-, or position-only.	No targeted change; complete-process re-evaluation not performed.
1.3.4	Orientation (AA)	Applicable	Responsive workflow states	Viewport and orientation review	Not fully evaluated: no session-level orientation or device record survives.	Responsive source retained; portrait and landscape retest remains required.
1.3.5	Identify Input Purpose (AA)	Not applicable	Provider and record selections	Form-purpose inspection	Not applicable: the evaluated form did not collect user-personal input covered by the input-purpose taxonomy.	Applicability must be reconfirmed if personal-data fields are added.
1.4.1	Use of Color (A)	Applicable	Statuses, selections, and controls	Visual/manual inspection	No issue observed: retained review did not identify a color-only cue.	No targeted change; complete-process re-evaluation not performed.
1.4.2	Audio Control (A)	Not applicable	All enumerated states	Content inventory	Not applicable: no automatically playing audio.	No audio added; applicability must be reconfirmed for future builds.
1.4.3	Contrast (Minimum) (AA)	Applicable	Recorded text/background pairs	Measured contrast review	No issue observed: 29/29 retained pairs met the stated 4.5:1 threshold (3:1 for large text).	Initial-state scan repeated; no full-state contrast re-evaluation.
1.4.4	Resize Text (AA)	Applicable	Enumerated responsive states	Manual zoom review	Not fully evaluated: a 200% check was summarized, but browser, viewport, and per-state evidence were not preserved.	No targeted change; repeat with preserved environment and state evidence.
1.4.5	Images of Text (AA)	Applicable	All enumerated states	Content inspection	No issue observed: retained review recorded real text rather than authored images of text.	No targeted change; complete-process re-evaluation not performed.

SC	Criterion	Applicability	Cycle 1 tested state	Method and state provenance	Cycle 1 result	Cycle 2 status
1.4.10	Reflow (AA)	Applicable	Dashboard and workflow at 320 CSS pixels	Manual responsive inspection (Cycle 1 summary); Cycle 2 narrow-viewport inspection with retained screenshots	Not fully evaluated: a 320-CSS-pixel check was summarized without a preserved state-by-state record.	Remediated and technically verified for the recorded narrow-viewport states: the status panel moves into normal flow and the earlier overlap was not reproduced; this is not exhaustive device coverage.
1.4.11	Non-text Contrast (AA)	Applicable	Controls, boundaries, and focus indicators	Measured/manual inspection	No issue observed: retained review recorded at least 3:1 for inspected components and focus indicators.	No full-state remeasurement after the Cycle 2 changes.
1.4.12	Text Spacing (AA)	Applicable	Enumerated text-bearing states	Manual spacing override	Not fully evaluated: the summary reports no clipping but no preserved per-state screenshots or settings.	Repeat with preserved spacing settings and workflow states.
1.4.13	Content on Hover or Focus (AA)	Applicable where additional content appears	Tooltips/disclosures in enumerated states	Keyboard, pointer, and code inspection	Not fully evaluated: no complete retained hover/focus-content test record.	No complete manual re-evaluation after the Cycle 2 changes.
2.1.1	Keyboard (A)	Applicable	Principal workflow controls in the evaluated build	Manual keyboard walkthrough (Cycle 1); A1–A12 screenshots and state reports (Cycle 2)	No issue observed: retained workflow review recorded keyboard operation.	Remediated and technically verified for the recorded principal workflow, audit dialog, revocation dialog, and FAQ controls; alternative-input testing remains outside scope.
2.1.2	No Keyboard Trap (A)	Applicable	Modal dialogs in the evaluated workflow	Manual Tab, Shift+Tab, Escape, and focus-restoration inspection; Cycle 2 states A7 and A8	No issue observed: no trap was recorded in the tested workflow.	Remediated and technically verified in the final Cycle 2 build; focus containment and restoration passed the recorded manual checks. No live screen-reader test was performed.
2.1.4	Character Key Shortcuts (A)	Not applicable	All enumerated states	Code and interaction inspection	Not applicable: no single-character shortcuts were identified.	No shortcuts added; applicability must be reconfirmed for future builds.
2.2.1	Timing Adjustable (A)	Minimum wait not applicable; real authentication timer not evaluated	Real timed authentication only; the minimum-wait countdown state excluded	Workflow and code inspection	Not fully evaluated: the 60-second mechanism is a minimum wait with no post-wait completion deadline, so it is not a time limit under this SC; timed authentication was not assessed.	No pass or failure is assigned to the countdown under this criterion. Real timed authentication still requires evaluation.
2.2.2	Pause, Stop, Hide (A)	Potentially applicable	Visible 60-second countdown and other updating content	Manual and code inspection	Not fully evaluated: no retained assessment establishes whether the visible auto-updating countdown requires or provides a pause, stop, or hide mechanism.	Countdown remains; criterion-specific manual evaluation is outstanding.
2.3.1	Three Flashes or Below Threshold (A)	Applicable	Animations and state changes	Visual and code inspection	No issue observed: no flashing content was recorded.	Reduced-motion handling retained; full-state recheck not performed.
2.4.1	Bypass Blocks (A)	Applicable	Initial page and revealed sections	Keyboard/code inspection	No issue observed: a skip-to-main link was recorded.	No targeted change; full-state keyboard recheck pending.
2.4.2	Page Titled (A)	Applicable	Document/page state	DOM inspection	No issue observed: a descriptive page title was recorded.	No targeted change.

SC	Criterion	Applicability	Cycle 1 tested state	Method and state provenance	Cycle 1 result	Cycle 2 status
2.4.3	Focus Order (A)	Applicable	Workflow, dialog, and error states	Manual keyboard and focus inspection (Cycle 1); Cycle 2 states A2, A4–A8, and A10–A12 with retained screenshots	No issue observed: no illogical focus order was recorded.	Remediated and technically verified: dashboard and wizard heading focus, countdown availability, success focus, dialog containment/restoration, and error focus passed the recorded manual checks.
2.4.4	Link Purpose (In Context) (A)	Applicable	Navigation, audit, and explorer links	Content and code inspection	No issue observed: retained review recorded descriptive link purpose in context.	Dynamic links require fresh rendered-state inspection.
2.4.5	Multiple Ways (AA)	Applicable to page sets	Navigation and direct actions	Structure inspection	Not fully evaluated: retained evidence does not establish the page-set scope or all navigation mechanisms.	Reassess if deployed as a multi-page product.
2.4.6	Headings and Labels (AA)	Applicable	Page regions, forms, and wizard	Content/manual inspection	No issue observed: retained review recorded descriptive headings and labels.	Affected wizard headings, dialog labels, and contextual control names were technically verified in the recorded Cycle 2 states; a complete criterion-wide evaluation is not claimed.
2.4.7	Focus Visible (AA)	Applicable	Enumerated keyboard states	Manual keyboard walkthrough	No issue observed: visible focus was recorded in the tested workflow.	Visible focus was verified for the recorded principal keyboard workflow and dialog states; exhaustive coverage across every responsive state is not claimed.
2.4.11	Focus Not Obscured (Minimum) (AA)	Applicable	Sticky content, overlays, and scrolled workflow states	Manual keyboard/viewport inspection	Not fully evaluated: no preserved test demonstrates that focused components remained at least partially visible in every state and viewport.	Requires a fresh keyboard test across overlays, scroll positions, and responsive viewports.
2.5.1	Pointer Gestures (A)	Not applicable	All scripted controls	Code and interaction inventory	Not applicable: no multipoint or path-based gesture was required.	No gesture control added; applicability must be reconfirmed for future builds.
2.5.2	Pointer Cancellation (A)	Applicable	Buttons, links, accordions, and modal dismissals	Pointer and code inspection	Not fully evaluated: no preserved down-event, up-event, abort, or undo test record.	Current controls use click/button activation; manual pointer-cancellation retest pending.
2.5.3	Label in Name (A)	Applicable	“Remove Access” and other visibly labelled controls	Accessible-name and visible-label comparison	No issue observed: the visible string “Remove Access” was contained in the control name; the separate lack of provider context is recorded under 3.3.2 and 4.1.2.	Remediated and technically verified for the Cycle 2 removal controls: the accessible name retains the visible “Remove Access” text and adds provider and record context.
2.5.4	Motion Actuation (A)	Not applicable	All scripted controls	Input-mechanism inventory	Not applicable: no device-motion or user-motion operation was required.	No motion input added; applicability must be reconfirmed for future builds.
2.5.7	Dragging Movements (AA)	Not applicable	All scripted controls	Input-mechanism and code inventory	Not applicable: no functionality required a dragging movement.	No dragging interaction added; applicability must be reconfirmed for future builds.
2.5.8	Target Size (Minimum) (AA)	Applicable	Record controls, explorer link, and FAQ controls	Rendered bounding-box measurement and screenshots; Cycle 2 states A3, A6, and A9	Issue observed: compact controls were flagged for review in the evaluated build.	Remediated and technically verified for the affected controls: checkboxes and the explorer link have minimum 24-by-24-CSS-pixel target dimensions and the recorded manual measurements passed.
3.1.1	Language of Page (A)	Applicable	Document root	DOM inspection	No issue observed: the page language was set.	No targeted change.

SC	Criterion	Applicability	Cycle 1 tested state	Method and state provenance	Cycle 1 result	Cycle 2 status
3.1.2	Language of Parts (AA)	Not applicable	All enumerated content	Content inventory	Not applicable: no passage in another human language was recorded.	Applicability must be reconfirmed if multilingual content is added.
3.2.1	On Focus (A)	Applicable	Focusable controls	Manual keyboard walkthrough	No issue observed: no context change on focus was recorded.	Full-state retest pending.
3.2.2	On Input (A)	Applicable	Provider/record selections and controls	Manual interaction walkthrough	No issue observed: no unexpected context change on input was recorded.	Full-state retest pending.
3.2.3	Consistent Navigation (AA)	Applicable to repeated navigation	Revealed workflow sections	Structural/manual inspection	No issue observed: retained review recorded consistent navigation.	Deployment as a page set requires a fresh consistency test.
3.2.4	Consistent Identification (AA)	Applicable	Repeated controls and components	Content/manual inspection	No issue observed: repeated components were recorded as consistently identified.	Dynamic components require fresh rendered-state inspection.
3.2.6	Consistent Help (A)	Applicable where help is repeated	FAQ/help access across workflow states	Cross-state structural inspection	Not fully evaluated: the retained materials do not document a criterion-specific consistency check for help placement and order.	Current source retains the FAQ/help region; complete cross-state evaluation remains required.
3.3.1	Error Identification (A)	Applicable	Record-loading, audit-loading, and transaction-failure presentations	Manual inspection (Cycle 1); verification-harness invocation, standalone axe-core, screenshot, visible-alert, and programmatic-announcement inspection in Cycle 2 states A10–A12	No issue observed: retained review recorded identified and announced errors in inspected states.	Remediated and technically verified for the recorded presentations; the retained files are listed in S5.2. Dynamic RPC-loss failure injection remains outside scope.
3.3.2	Labels or Instructions (A)	Applicable	Record list and revocation controls	Manual accessible-name/content inspection (Cycle 1); standalone axe-core in Cycle 2 states A3 and A8	Issue observed: the “Remove Access” control lacked provider context in the evaluated build.	Remediated and technically verified: provider and record context appears in Cycle 2 removal-control names and the accessible revocation dialog was inspected.
3.3.3	Error Suggestion (AA)	Applicable where suggestions are known	Validation and transaction error states	Manual failure-state inspection	Not fully evaluated: no complete retained set of error and correction-guidance states.	Fresh failure-injection and guidance review required.
3.3.4	Error Prevention (Legal, Financial, Data) (AA)	Applicable to consent changes	Grant review, confirmation, cancellation, and revocation	Complete-process and code inspection	Not fully evaluated: two-stage confirmation and cancellation provide partial evidence, but complete-process prevention testing was not preserved.	Cancellation and state wording were improved; complete-process re-evaluation remains required.
3.3.7	Redundant Entry (A)	Potentially applicable	Multi-step grant workflow	Complete-process manual inspection	Not fully evaluated: no retained criterion-specific record establishes whether previously entered information was requested again or automatically populated.	Current wizard retains selections when navigating between stages; complete-process retest pending.
3.3.8	Accessible Authentication (Minimum) (AA)	Applicable to real authentication	Simulated fingerprint unlock	Manual and code inspection	Not fully evaluated: fingerprint interaction was simulated and did not constitute a real authentication mechanism.	Real WebAuthn/FIDO2 or alternative authentication remains unimplemented and unevaluated.

SC	Criterion	Applicability	Cycle 1 tested state	Method and state provenance	Cycle 1 result	Cycle 2 status
4.1.2	Name, Role, Value (A)	Applicable	Record, dialog, and FAQ controls	Manual accessibility-tree and code inspection (Cycle 1); standalone axe-core and screenshot inspection in Cycle 2 states A3 and A7–A9	Issue observed: the evaluated “Remove Access” control lacked contextual accessible-name information.	Remediated and technically verified: corrected controls and both dialogs expose names, roles, and states in the inspected build; live screen-reader confirmation remains outstanding.
4.1.3	Status Messages (AA)	Applicable	Wizard, confirmation, and error states	Live-region/code inspection (Cycle 1); manual focus checks, standalone axe-core, and retained screenshots in Cycle 2 states A2, A4–A6, and A10–A12	Issue observed: wizard step transitions were incompletely announced in the evaluated build.	Remediated and technically verified within the recorded browser scope: transition focus, confirmation availability, success, and visible/programmatic error messages were inspected; live NVDA/VoiceOver evaluation remains required.

Evidence boundary. The matrix preserves the Cycle 1 classifications and identifies Cycle 2 source changes separately. The Cycle 1 commit and environment were not archived (L4). The 15 July 2026 Lighthouse and axe-core reports cover only the Cycle 2 initial rendered state. The final 15 July standalone axe-core 4.10.0 layer covers the 12 states listed in S5.2; keyboard, focus, target-size, error-presentation, and reflow evidence is manual/browser evidence recorded separately. No layer constitutes a WCAG 2.2 conformance evaluation or live screen-reader test. Initial-state reports are deposited as `results/wcag/lighthouse_report_2026-07-15.json` and `results/wcag/axe_report_2026-07-15.json`; final state reports and screenshots are under `results/wcag/final-state-verification/`.

S6. Final Cycle 2 Contract Technical Verification

The final Hardhat suite was executed on 15 July 2026 using solc 0.8.26 and Hardhat Chain ID 1337. It contained 28 tests, all of which passed. The retained output covers authorized and unauthorized audit logging, owner-controlled gateway administration, zero-address rejection, owner and non-owner pause administration, blocked grant initiation and confirmation while paused, continued cancellation and revocation while paused, and normal operation after unpausing, together with the retained lifecycle and access-state cases. The Flask gateway rejects state-changing requests unless exactly one record identifier is supplied before transaction submission. Dedicated dynamic RPC-loss failure-injection testing remains outstanding.

Slither 0.11.5 was rerun on the final Cycle 2 `HealthAccessControl.sol` using 101 detectors. Canonical retained filenames are listed separately to avoid ambiguity:

- `results/slither-report.json`;
- `results/slither-summary.txt`; and
- `results/hardhat-test-output.txt`.

Severity	Count	Detector(s)	Resolution
High	0	—	No high-severity detector output.
Medium	0	—	No medium-severity detector output.
Low	6	<code>timestamp</code>	<code>block.timestamp</code> dependence is intrinsic to the documented minimum-wait and expiry design; permitted miner/validator variation remains a boundary risk (L9).
Informational	35	<code>naming-convention</code>	Parameter names use a leading underscore; this is a style finding with no functional effect.

The final source restricts access-attempt logging to the owner or authorized gateway and applies pause enforcement to grant initiation and confirmation while preserving cancellation and revocation. Static analysis is a floor, not a substitute for professional audit; property-based fuzzing, dynamic adversarial analysis, dedicated gateway failure injection, dependency review, multi-node testing, and independent audit remain outstanding (Limitation L9).

S7. Targeted Narrative Evidence-Gathering Notes

The related-work synthesis was conducted to motivate the research problem, identify relevant accessibility and consent-interface concerns, and position BCHAIN-ACCESS against closely related work. It was not designed as a systematic literature review, scoping review, bibliometric analysis, or exhaustive evidence synthesis.

Relevant work was identified from IEEE Xplore, ACM Digital Library, SpringerLink, and PubMed using combinations of terms related to blockchain, healthcare, consent, HCI, usability, accessibility, and assistive technology. Seminal earlier studies were retained where they defined relevant blockchain-healthcare systems or HCI concepts.

Studies were considered relevant when they addressed blockchain healthcare together with consent, interface design, usability, accessibility, or patient-controlled access. Infrastructure-only and cryptocurrency-only studies were outside the focused synthesis. The literature was used qualitatively for problem formulation and design reasoning; no prevalence estimates, exhaustive-coverage claims, or systematic-review conclusions are made.

S8. Evaluation Detail, Data Provenance, and Exclusions

S8.1 Documentation-Based Heuristic Evidence Matrix

The matrix below records categorical evidence from the documentation review. It does not represent live-system observation, user impact, severity, or comparative performance.

Nielsen Heuristic	MedRec	Patientory	BurstIQ
H1: Visibility of System Status	Concern evidenced	Concern evidenced	Concern partially evidenced
H2: Match System & Real World	Concern evidenced	Concern partially evidenced	Concern evidenced
H3: User Control & Freedom	Concern evidenced	Concern evidenced	Concern evidenced
H4: Consistency & Standards	Concern partially evidenced	Concern partially evidenced	Concern partially evidenced
H5: Error Prevention	Concern evidenced	Concern evidenced	Concern evidenced
H6: Recognition over Recall	Concern evidenced	Concern partially evidenced	Concern evidenced
H7: Flexibility & Efficiency	Concern partially evidenced	Concern partially evidenced	Concern partially evidenced
H8: Aesthetic & Minimalist Design	Concern partially evidenced	Concern not evidenced	Concern partially evidenced
H9: Error Recovery	Concern evidenced	Concern evidenced	Concern evidenced
H10: Help & Documentation	Concern evidenced	Concern partially evidenced	Concern evidenced

S8.2 Evaluator-Level Descriptive Metadata

ID	Reported background	Yrs.	HCI	Blockchain	Accessibility
E1	Accessibility Researcher	7	Expert	Intermediate	Expert
E2	UX Designer & Consultant	5	Expert	Novice	Expert
E3	HCI Research Scientist	6	Expert	Intermediate	Advanced
E4	Software Engineering Faculty	5	Expert	Intermediate	Advanced
E5	Senior Frontend Developer	8	Expert	Intermediate	Expert
E6	Accessibility Specialist (CPACC, IAAP)	9	Expert	Novice	Expert
E7	UX Researcher (PhD HCI, Health Informatics)	7	Expert	Intermediate	Advanced
E8	Senior Frontend Engineer (Accessibility Advocate)	10	Expert	Intermediate	Expert
E9	Occupational Therapist (Digital Health)	8	Expert	Novice	Advanced
E10	Assistive Technology Specialist	9	Expert	Intermediate	Expert

Expertise labels and credential strings reproduce the retained evaluator summary. Their definitions, assignment method, and independent verification were not preserved, so they are descriptive metadata rather than independently established grades.

S8.3 Coarse Heuristic Rating Matrix

Excluded from the reported results (main paper, ED2 and Limitation L6). Reproduced unaltered for transparency: all ten evaluators returned an identical rating vector, which cannot be shown to reflect ten independent observations. No claim in the paper rests on these values.

ID	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	Mean
E1	0	0	0	0	0	0	0	0	1	0	0.1
E2	0	0	0	0	0	0	0	0	1	0	0.1
E3	0	0	0	0	0	0	0	0	1	0	0.1
E4	0	0	0	0	0	0	0	0	1	0	0.1
E5	0	0	0	0	0	0	0	0	1	0	0.1
E6	0	0	0	0	0	0	0	0	1	0	0.1
E7	0	0	0	0	0	0	0	0	1	0	0.1
E8	0	0	0	0	0	0	0	0	1	0	0.1
E9	0	0	0	0	0	0	0	0	1	0	0.1
E10	0	0	0	0	0	0	0	0	1	0	0.1

The uniform pattern gives 100% raw agreement but a degenerate marginal distribution. A chance-corrected coefficient is therefore not interpreted.

S8.4 Expert Task-Workbook Results

Timings below are reported descriptively in the main paper. The completion and error fields are excluded from the reported results (main paper, ED3 and Limitation L6): they are uniform across all ten evaluators and “error” is not operationally defined. See S8.8.

Task	Workbook-reported completion	Mean active time (s)	SD	Errors	Difficulty
T1: Interface entry	100%	6.8	3.9	0.0	1.5
T2: View Record	100%	4.0	1.5	0.0	1.6
T3: Check Access	100%	6.3	2.2	0.0	1.5
T4: Grant Access	100%	14.8	5.9	0.0	2.4
T5: Review	100%	10.4	3.9	0.0	1.8
T6: Confirm	100%	17.3	6.3	0.0	2.4
T7: Revoke	100%	7.0	2.8	0.0	1.7
T8: FAQ	100%	10.9	5.2	0.0	2.1
Overall	100%	9.7	5.9	0.0	1.9

Completion values are reproduced from the consolidated workbook and cannot be independently reconstructed from the task CSV. Difficulty is rated 1 (very easy)–5 (very difficult). The 17.3-second T6 value records active interaction after the separate 60-second minimum wait.

S8.5 SUS Scores and Scoring

Excluded from the reported results (main paper, ED4 and Limitation L6). Reproduced for transparency only: the respondents were unblinded non-target evaluators inspecting an artifact built around the principles they were shown, so these scores cannot support an accessibility or usability claim.

For the standard ten-item SUS, odd-item contributions are $X_i - 1$ and even-item contributions are $5 - X_i$; the sum is multiplied by 2.5:

$$\text{SUS} = \left(\sum_{i=1,3,5,7,9} (X_i - 1) + \sum_{i=2,4,6,8,10} (5 - X_i) \right) \times 2.5.$$

Metric	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	Mean
SUS Score	87.5	90.0	82.5	90.0	97.5	87.5	85.0	92.5	82.5	80.0	87.5

$SD = 5.27$; range 80.0–97.5. These values describe perceived usability within the expert sample only.

S8.6 Custom Questionnaire Item Results

Excluded from the reported results (main paper, ED5 and Limitation L6). Reproduced for transparency only: the instrument is author-designed with no reliability or validity evidence and was completed by the same unblinded non-target evaluators as S8.5.

Statement	Mean	SD
1. I found the navigation consistent and predictable.	4.8	0.4
2. I found the instructions and labels easy to understand.	4.4	0.5
3. I was able to access all features using only the keyboard.	4.5	0.5
4. I found the colour contrast adequate for reading.	4.8	0.4
5. I found the error messages helpful and clear.	4.2	0.4
6. I felt in control while using the system.	4.5	0.7
7. I would feel comfortable using this system independently.	4.1	0.6
8. I found the system forgiving if I made a mistake.	4.7	0.5
9. I understood the information presented to me.	4.0	0.5
10. I found the system responsive and fast enough.	4.4	0.5
Overall participant-level mean	4.44	0.36

Scale: 1 (Strongly Disagree) to 5 (Strongly Agree). The instrument is author-designed, and its reliability and validity have not been evaluated. Participant-level means ranged from 3.8 to 4.9.

S8.7 Hypothetical Workflow-to-Principle Mapping

This mapping expands the diagnosis-neutral traceability example in the main paper and shows how the reference artifact represents each principle.

Step	Workflow action	Principle(s)	Artifact representation	Evidence boundary
T1	Enter the interface	AAD	Simulated entry control	Production authentication not implemented
T2	View an MRI record	PTD	Record and metadata presentation	Semantic separation identified for follow-up
T3	Inspect provider access	PTD, ATC	Permission list and removal controls	Contextual names and permission controls remediated and technically retested
T4	Initiate a grant	PTD, CID	Provider selection and grant initiation	Single-record initiation and transition handling verified
T5	Review consequences	CID	Review page before confirmation	Wizard focus movement verified; cognitive content-density effectiveness remains target-user work
T6	Confirm after the minimum wait	CID, ATC	Delayed confirmation and status messaging	Countdown status and focus technically verified; acceptability remains empirically unresolved
T7	Revoke access	CID, PTD	Revocation and updated permission state	Accessible revocation dialog and focus behavior technically verified
T8	Open help content	PTD, ATC	Plain-language FAQ and accordion controls	FAQ target-size corrections technically verified

S8.8 Evaluation Data Provenance and Exclusions

The source package contains a consolidated ten-evaluator analysis workbook and exported CSV tables. The task CSV retains evaluator ID, task ID and name, time, error count, difficulty, and notes, but no row-level completion, assistance, restart, or incomplete-attempt field. It contains no timestamps or software telemetry. The collection packet does not preserve whether evaluators self-timed, entered values immediately after each task, or supplied values for later transcription; no observer sheet, screen recording, or session recording is deposited. “Error” is not operationally defined beyond the numeric field, the aggregate “without assistance” statement cannot be independently checked, and restart and incomplete-attempt rules were not retained. Task values and the consolidated workbook’s 100% completion figure are therefore workbook-reported expert-task data rather than independently observed performance measures.

The expert task times are recorded as active interaction time; T6 excludes the fixed 60-second system wait, so its 17.3-second mean corresponds to a minimum elapsed duration of 77.3 seconds. The custom questionnaire has not undergone reliability or validity evaluation, and participant-level rather than item-level dispersion is used when describing respondent variability. Recruitment channel, evaluator-author relationships, disability status, definitions and assignment of the “Advanced”/“Expert” labels, and credential verification were not preserved. Documented WhatsApp consent records are retained privately because they contain participant-identifying information and are not included in the deposited research package.

Per-session browser, operating-system, screen-reader version, zoom/reflow configuration, and alternative-input settings were not preserved. E10’s comments identify NVDA use without a version. The manual findings are therefore diagnostic expert observations rather than a reproducible assistive-technology compatibility test.

The consolidated workbook contains heuristic-level means for a finer post-hoc re-rating but does not contain its scale anchors, instructions, administration time, whether prior answers were visible, or the complete per-evaluator matrix. Consequently, that re-rating is excluded from the principal results. The baseline source package likewise lacks separate pre-consensus contingency tables needed to recompute weighted ordinal and categorical agreement; the pooled $\kappa = 0.74$ is withdrawn and no replacement reliability coefficient is claimed.

S9. Detailed Implementation Scope

This table expands the condensed implementation boundary in the main paper. “Implemented” refers to the reference development artifacts, not a production deployment.

Component	UI-layer prototype	Local-development backend	Production requirement
Blockchain layer	Integrated UI could use the local EVM or explicit fallback; blockchain behavior was not an accessibility-evaluation outcome	<code>HealthAccessControl.sol</code> on a local Hardhat EVM via <code>Web3.py</code> , with an in-memory fallback ledger	Governed distributed deployment with documented finality and operational assumptions

Component	UI-layer prototype	Local-development backend	Production requirement
Smart contracts	Contract behavior was not assessed by the expert accessibility instrument	Solidity 0.8.26 contract deployed and integration-tested on a local Hardhat node	Independently audited contracts under production governance
Cryptography	Cryptographic correctness was not assessed by the expert instrument	Seeded SHA-256 identifiers, EVM nonces, and Web3.py signing with a Hardhat development key; no production record encryption or salt store	Deployment-appropriate hashing, encryption, user key management, and HSM/KMS-backed secrets
Wallet integration	No user-controlled wallet or key management	Patient wallet derived from a Hardhat test account; backend signs development transactions	Accessible user-controlled wallet, key derivation, recovery, and secure storage
Data persistence	The Cycle 2 interface uses API-managed state; there is no separate browser-memory Cycle 2 build	In-memory Python fallback state; seeded identifiers registered on startup when the local EVM is available	Governed on-chain records plus secured, encrypted off-chain clinical storage
Authentication	Simulated fingerprint control only	Wallet-address checks and <code>onlyPatient()</code> contract restriction; no real biometric authentication	Evaluated WebAuthn/FIDO2 or other accessible identity-assurance workflow
Representative access	Not implemented	Not implemented	Jurisdiction-specific authority, scope, expiry, revocation, identification, and abuse safeguards
API gateway	Frontend calls the reference REST API	Flask REST API connects the frontend to the Hardhat EVM	Authenticated, rate-limited, monitored, TLS-protected production service with durable storage
Transaction finality	Displays local-EVM hashes or explicit fallback identifiers; mode was not an accessibility outcome	Local-EVM transactions confirmed when available; fallback behavior is separately identified	Deployment-specific confirmation, reorganization, outage, and retry handling
Accessibility evidence	Selected Cycle 1 implementation issues addressed and technically verified in Cycle 2; 12 standalone axe-core state reports and screenshots retained	No live screen-reader or disabled-user validation	Whole-artifact WCAG 2.2, assistive-technology, alternative-input, and disabled-participant evaluation with production authentication and wallet components
Security assurance	No independent assurance	Final 28-test suite, Slither, authorized logger, pause policy, and single-record API enforcement; no independent audit, fuzzing, or adversarial testing	Independent audit, property-based fuzzing, failure injection, adversarial testing, dependency migration, and multi-node evaluation

The UI prototype was the subject of the ten-expert feasibility assessment. The local-EVM backend illustrates technical feasibility only. Neither component is production-ready.

S9.1 Condensed Implementation Status

Component	Implemented	Retained evidence	Production-ready
UI prototype	Yes	Cycle 1 expert-inspected workflow plus final 12-state axe-core and manual-browser verification (Cycle 2)	No
Local-EVM backend	Yes	Final 28-test suite, local integration verification, and Slither 0.11.5 static analysis	No
Real biometric authentication	No	Simulated entry only	No
Guardian delegation	No	Framework recommendation	No
Production record storage	No	Development identifiers only	No
Independent security audit	No	Static analysis only	No

S9.2 Cycle 1 Artifact Metadata

Required item	Available record
Frontend version	Cycle 1 commit hash or release tag not archived.
Backend version	Cycle 1 commit hash or release tag not archived. The Cycle 2 code/evidence snapshot is commit <code>360eaa2a26b2f2b4fef98c0667f87db09e5e5120</code> .
Evaluation date	Exact session dates not recorded. The retained Cycle 1 automated reports are dated 29 June and 12 July 2026, which places the cycle before the 15 July Cycle 2 verification.
Backend mode by session	Not logged; the interface could use the local EVM when available or the explicit fallback otherwise.
Deployment URL	The Cycle 1 reference URL was not archived as a versioned release and is not treated as a stable artifact identifier.
Evaluator environment	Per-session browser, operating system, and assistive-technology versions not preserved; E10 comments identify NVDA without version.

These missing identifiers prevent exact reconstruction of the Cycle 1 artifact. The deposited repository is the Cycle 2 build.

Artifact availability. Zenodo DOI <https://doi.org/10.5281/zenodo.21345578> identifies the earlier v1.0.0 release published on 13 July 2026. Inspection of the Zenodo record shows that its single archive predates the 15 July verification and therefore does not contain the final 28-test output, 15 July Slither JSON, live axe evidence, or the Cycle 2 verification report. It is not cited as the archive of the final evidence package. Before publication, a new versioned Zenodo release must deposit the exact Cycle 2 source and these retained reports; its version DOI should then replace this provisional availability statement.

S10. Final Cycle 2 Engineering Verification

Item	Retained verification record
Date and source identity	15 July 2026. Cycle 2 code/evidence commit <code>360eaa2a26b2f2b4fef98c0667f87db09e5e5120</code> .
Accessibility tools and states	Standalone axe-core 4.10.0 evaluated landing, dashboard, record selection, review, countdown, confirmation success, audit-dialog open, revocation-dialog open, expanded FAQ, record-loading failure, audit-loading failure, and transaction failure. Lighthouse 13.4.0 with embedded axe-core 4.12.1 remains a separate Cycle 2 initial-state artifact.
Standalone axe-core result	Twelve retained reports recorded zero violations; five reports contain incomplete items retained for manual review. The intermediate axe-core 4.8.2 dialog finding is preserved separately as pre-final-fix evidence.
Manual browser checks	Principal keyboard workflow, wizard-transition focus, audit and revocation dialog focus containment/restoration and Escape handling, affected target dimensions, visible/programmable error presentation, and 320-CSS-pixel reflow passed the recorded checks.
Contract tests	The final Hardhat suite contained 28 tests, all of which passed; retained output includes authorized logger, gateway administration, pause-policy, cancellation/revocation-while-paused, and post-unpause cases.
Static analysis	Slither 0.11.5: six low timestamp findings and 35 informational naming-convention findings; no high- or medium-severity detector findings were reported.
Final security source status	Logger restricted to owner/authorized gateway; initiation and confirmation blocked while paused with cancellation and revocation retained; state-changing API requests require exactly one record.
Residual limitations	No live screen-reader, alternative-input, or disabled-participant evaluation; no whole-artifact WCAG conformance determination; no property-based fuzzing, dynamic failure injection, adversarial testing, multi-node evaluation, production key management, or independent audit.
Evidence inventory	Twelve axe JSON files and screenshots under results/wcag/final-state-verification/ ; results/hardhat-test-output.txt ; results/slither-report.json ; results/slither-summary.txt ; results/gas-report.txt ; Lighthouse JSON; verification matrices; source; README; environment example; artifact manifest; and the final verification report.